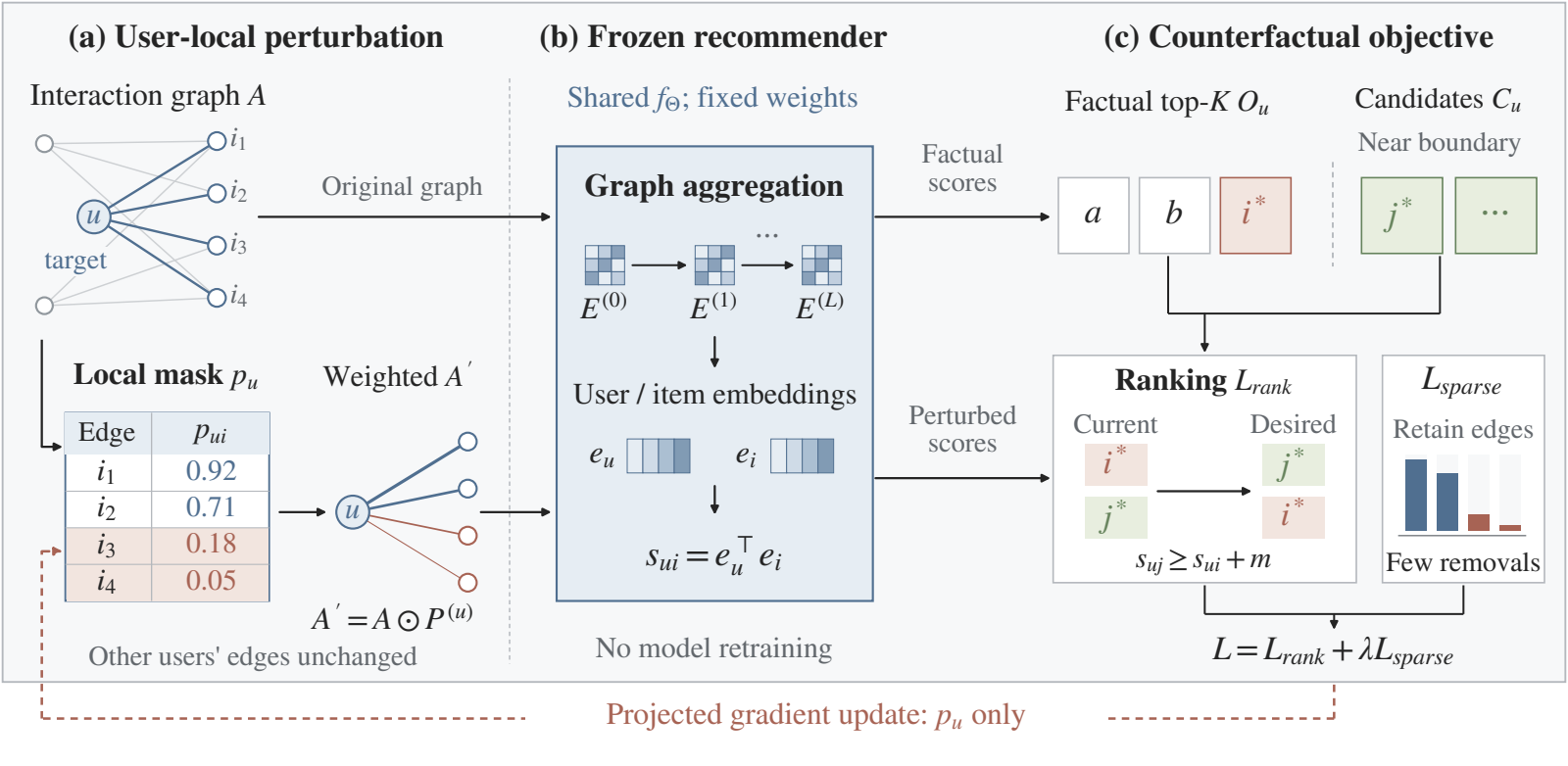
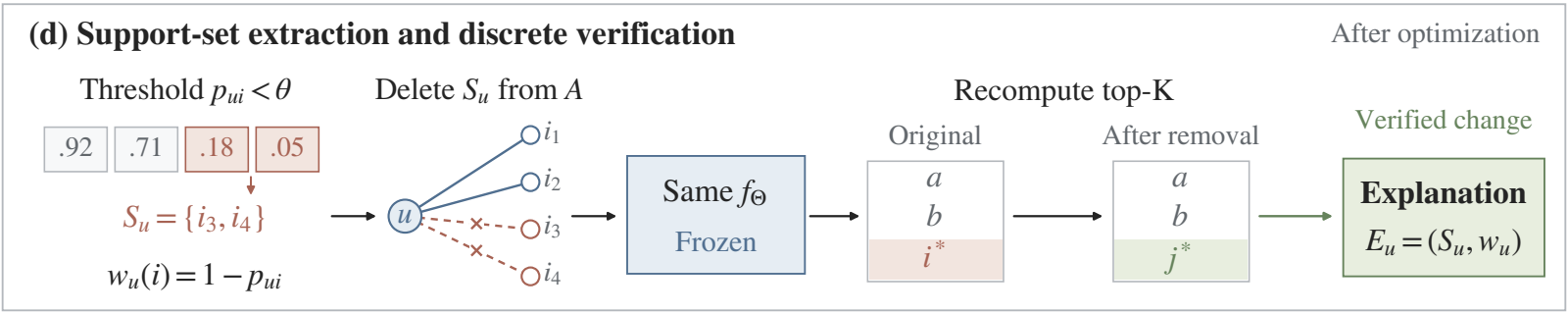

(a) User-local perturbation
(b) Frozen recommender
(c) Counterfactual objective
Interaction graph A
i_1
i_2
i_3
i_4
u
target
Original graph
Shared f_Θ; fixed weights
Graph aggregation
$E^{(0)}$
$E^{(1)}$
$E^{(L)}$
User / item embeddings
e_u
e_i
$s_{ui} = e_u^\top e_i$
Factual scores
Factual top-K O_u
Candidates C_u
Near boundary
a
b
i^*
j^*
Local mask p_u
Edge
p_{ui}
i_1 0.92
i_2 0.71
i_3 0.18
i_4 0.05
Weighted A'
$A' = A \odot P^{(u)}$
Other users' edges unchanged
No model retraining
Perturbed scores
Ranking L_{rank}
Current
Desired
$s_{uj} \geq s_{ui} + m$
L_{sparse}
Retain edges
Few removals
$L = L_{rank} + \lambda L_{sparse}$
Projected gradient update: p_u only


(d) Support-set extraction and discrete verification
After optimization
Threshold $p_{ui} < \theta$
.92
.71
.18
.05
$S_u = \{i_3, i_4\}$
$w_u(i) = 1 - p_{ui}$
Delete S_u from A
u
i_1
i_2
i_3
i_4
Same f_Θ
Frozen
Recompute top-K
Original
After removal
a
b
i^*
j^*
Verified change
Explanation
$E_u = (S_u, w_u)$